

# 互对比标定使用说明

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## 整体介绍

根据激光雷达对比标定软件的整体功能（如图1所示），可以将程序的运行逻辑分为“数据转换→自对比/互对比→输出结果”三个部分。三个部分的流程均通过配置文件（config file）控制，其中配置文件为 [YAML](#) 标记语言格式。

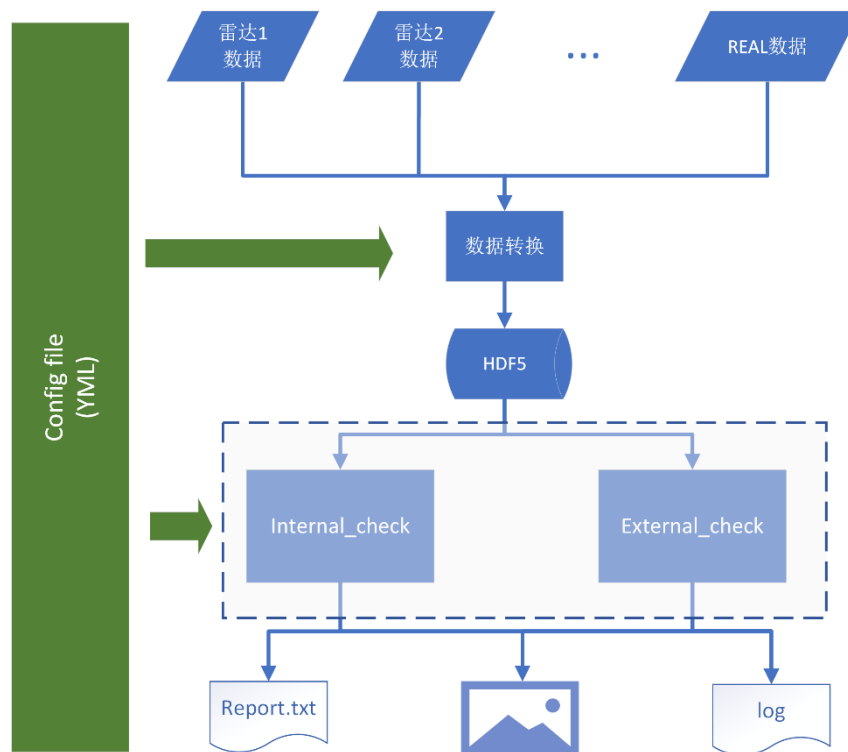


图1 激光雷达对比标定软件整体架构图

## 数据准备

因为 Matlab 对中文路径识别存在问题，因此需要讲所有的数据放在英文路径下。

| D:\Data\CMA_Lidar_Comparison\externalChk |                  |     |
|------------------------------------------|------------------|-----|
| 名称                                       | 修改日期             | 类型  |
| AW                                       | 2021/10/1 20:50  | 文件夹 |
| DS                                       | 2021/9/29 20:27  | 文件夹 |
| DS1                                      | 2021/10/10 14:45 | 文件夹 |
| GB                                       | 2021/9/29 13:58  | 文件夹 |
| HR                                       | 2021/9/29 14:02  | 文件夹 |
| L7                                       | 2021/9/29 13:51  | 文件夹 |
| REAL                                     | 2021/9/29 18:00  | 文件夹 |
| SS                                       | 2021/9/30 10:53  | 文件夹 |
| SY                                       | 2021/9/30 10:53  | 文件夹 |
| WH1                                      | 2021/9/30 13:30  | 文件夹 |
| WH2                                      | 2021/9/30 13:30  | 文件夹 |
| ZK                                       | 2021/9/29 13:55  | 文件夹 |

图 2 2021 年 9 月南郊对比实验互对比数据存储样例。图中每个文件夹代表一个厂家的雷达系统互对比数据

## 数据读取配置

在进行互对比时，我们需要将进行对比的数据从原始格式转换成统一的 [HDF5](#) 格式，因此在配置读取数据时需要配置多个雷达的读取设置，如下为读取 2021 年 09 月北京南郊对比标定雷达的设置（配置文件路径为 configBkp/Beijing/external\_check\_config.yml）：

```
#####
# data loader configuration
# Load lidar data to a specified directory (defined by `dataSavePath`)
#####
dataLoaderCfg:
  lidarList: ['AW', 'SY', 'L7', 'DS', 'DS1', 'SS', 'GB', 'ZK', 'WH1', 'WH2', 'REAL', 'HR']
  AW: # Ai wo si
    dataPath: D:\Data\CMA_Lidar_Comparison\externalChk\AW
    dataFilenamePattern: .*Lidar.* # all files
    dataFormat: 3
    chTag: ['532p', '532s']
    nMaxBin: 1800 # number of bins
    flagFilenameTime: true
  SY: # Shan yi suo
    dataPath: D:\Data\CMA_Lidar_Comparison\externalChk\SY
    dataFilenamePattern: .*Lidar.* # all files
    dataFormat: 3
    chTag: ['532p', '532s', '355e', '607', '1064e']
    nMaxBin: 1600 # number of bins
    flagFilenameTime: true
  L7: # 704
    dataPath: D:\Data\CMA_Lidar_Comparison\externalChk\L7
    dataFilenamePattern: .*Lidar.* # all files
    dataFormat: 3
    chTag: ['355p', '355s', '532p', '532s', '387', '407', '607', '1064e']
```

```
nMaxBin: 1800    # number of bins
flagFilenameTime: true
DS:    # Dasun
    dataPath: D:\Data\CMA_Lidar_Comparison\externalChk\DS
    dataFilenamePattern: .*Lidar.*
    dataFormat: 3
    chTag: ['1064p', '1064s']
    nMaxBin: 1250
DS1:    # Dasun1
    dataPath: D:\Data\CMA_Lidar_Comparison\externalChk\DS1
    dataFilenamePattern: .*Lidar.*
    dataFormat: 3
    chTag: ['1064e']
    nMaxBin: 1250
    flagFilenameTime: true
SS:    # Qianhai
    dataPath: D:\Data\CMA_Lidar_Comparison\externalChk\SS
    dataFilenamePattern: .*Lidar.*
    dataFormat: 3
    chTag: ['532p', '532s']
    nMaxBin: 1000
    nBin: 8000
GB:    # Zhongke Guang bo
    dataPath: D:\Data\CMA_Lidar_Comparison\externalChk\GB
    dataFilenamePattern: .*Lidar.*
    chTag: ['532p', '532s']
    dataFormat: 3
    nMaxBin: 2000
ZK:    # Wu xi zhong ke
    dataPath: D:\Data\CMA_Lidar_Comparison\externalChk\ZK
    dataFilenamePattern: .*Lidar.*
    dataFormat: 3
    chTag: ['532p', '532s']
    nMaxBin: 2000
WH1:
    dataPath: D:\Data\CMA_Lidar_Comparison\externalChk\WH1
    dataFilenamePattern: .*    # all files
    dataFormat: 1
    chTag: ['1064e']
    nMaxBin: 2048    # number of bins
WH2:
    dataPath: D:\Data\CMA_Lidar_Comparison\externalChk\WH2
    dataFilenamePattern: .*    # all files
    dataFormat: 2
    chTag: ['1064e']
    nMaxBin: 1300    # number of bins
REAL:
    dataPath: D:\Data\CMA_Lidar_Comparison\externalChk\REAL
```

```
dataFilenamePattern: .* # all files
dataFormat: 5
chTag: ['532sh', '532ph', '532sl', '532pl', '607l', '607h']
nMaxBin: 2300 # number of bgBins
HR:
dataPath: D:\Data\CMA_Lidar_Comparison\externalChk\HR
dataFilenamePattern: .*
dataFormat: 3
chTag: ['532p', '532s']
nMaxBin: 1900
```

设置完配置文件后，可以在 Matlab 命令行中运行读取数据的命令将雷达数据转换成 [HDF5](#) 格式。首先打开 Matlab，然后将当前目录导航到代码目录中，如图3所示

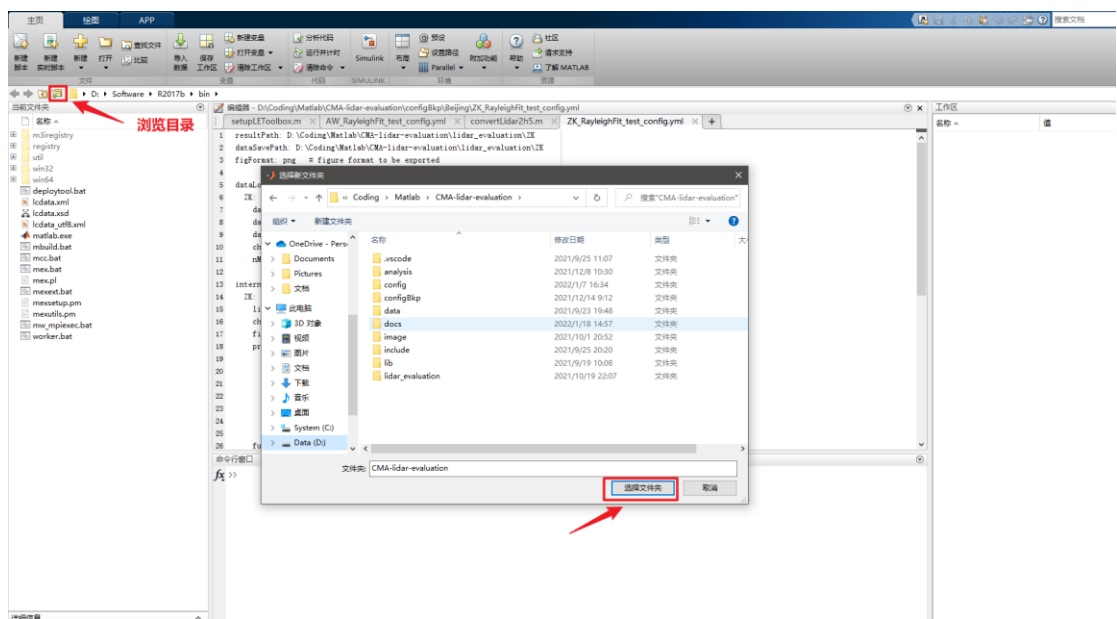


图 3 Matlab 中修改当前工作目录到对比软件代码目录中

然后在命令行中输入”setupLEToolbox”装载对比软件（或者点击 setupLEToolbox.m 文件打开，并单击运行），这里需要注意 Matlab 命令行大小写敏感，因此需要注意大小写一致。

装载完对比软件后，可以在命令行中看到一些输出结果。接下来可以运行如下指令转换多台雷达数据

```
LEMain('D:\Coding\Matlab\CMA-lidar-evaluation\configBkp\Beijing\external_check_test.yml', 'flagReadData', true)
```

```
命令窗口
In externalChk (line 63)
In LEMain (line 99)
[2022-01-18 15:16:15] Finish!
>> LEMain('D:\Coding\Matlab\CMA-lidar-evaluation\configBkp\Beijing\external_check_test.yml', 'flagReadData', true)
[2022-01-18 15:16:40] Start reading configurations!
[2022-01-18 15:16:40] Config file: D:\Coding\Matlab\CMA-lidar-evaluation\configBkp\Beijing\external_check_test.yml
[2022-01-18 15:16:40] Finish!
[2022-01-18 15:16:40] Start data conversion!
[2022-01-18 15:16:40] Convert lidar data for REAL
[2022-01-18 15:16:40] Reading REAL data.
[2022-01-18 15:16:41] 1184 data files were found!
[2022-01-18 15:16:41] Wait until reading finishes!
```

图 4 转换雷达数据

等待运行完成后，多台激光雷达数据会以.h5 格式的文件保存在 dataSavePath 中。

## 互对比距离修正信号

在读取完数据后，接下来可以通过设置互对比配置来完成互对比相关内容，这里以距离修正信号为例。

针对每台雷达需要先设置好雷达的预处理参数修正雷达的系统效应，预处理参数设置如下所示（配置文件路径为 configBkp/Beijing/external\_check\_config.yml，参考文件最后的内容）

```
#####
#
#                               external-check
#
# Below is the struct of `externalChkCfg`, which was used to inter-compare lidars.
#
#####
externalChkCfg:
  WH1:
    lidarNo: 12  # lidar number. (see ./docs/lidarList.md)
    chTag: ['1064e']
    fullOverlapHeight: 400  # minimum height with complete overlap. (m)
    overlapFile: ''
    hOffset: 0  # height offset. (m)
    tOffset: 0  # time offset. (min)
    deadTime: []  # deadtime (ns). If it's empty, deadtime correction is disabled.
    bgBins: [1500, 2000]  # [start index, stop index] for background correction
    nPretrigger: 15  # if nPretrigger < 0, move signal up
    bgCorFile: ''  # data file of dark measurement results
  WH2:
    lidarNo: 13  # lidar number. (see ./docs/lidarList.md)
    chTag: ['1064e']
    fullOverlapHeight: 200  # minimum height with complete overlap. (m)
    overlapFile: ''
    hOffset: 0  # height offset. (m)
    tOffset: 0  # time offset. (min)
    deadTime: []  # deadtime (ns). If it's empty, deadtime correction is disabled.
    bgBins: [1100, 1250]  # [start index, stop index] for background correction
```

```

nPretrigger: 0 # if nPretrigger < 0, move signal up
bgCorFile: '' # data file of dark measurement results
REAL:
lidarNo: 11 # lidar number. (see ./docs/lidarList.md)
chTag: ['532sh', '532ph', '532sl', '532pl', '607l', '607h']
fullOverlapHeight: 400 # minimum height with complete overlap. (m)
hOffset: 0 # height offset. (m)
tOffset: 0 # time offset. (min)
deadTime: [3.5, 3.5, 3.5, 3.5, 3.5, 30.4] # deadtime (ns). If it's empty, deadtime
correction is disabled.
bgBins: [1700, 2000] # [start index, stop index] for background correction
nPretrigger: 55 # if nPretrigger < 0, move signal up
bgCorFile: '' # data file of dark measurement results
overlapFile: ''
DS:
lidarNo: 9 # lidar number. (see ./docs/lidarList.md)
chTag: ['1064p', '1064s']
fullOverlapHeight: 400 # minimum height with complete overlap. (m)
hOffset: 0 # height offset. (m)
tOffset: 0 # time offset. (min)
deadTime: [] # deadtime (ns). If it's empty, deadtime correction is disabled.
bgBins: [1100, 1200] # [start index, stop index] for background correction
nPretrigger: 0 # if nPretrigger < 0, move signal up
bgCorFile: '' # data file of dark measurement results
overlapFile: ''
DS1:
lidarNo: 6 # lidar number. (see ./docs/lidarList.md)
chTag: ['1064e']
fullOverlapHeight: 400 # minimum height with complete overlap. (m)
hOffset: 0 # height offset. (m)
tOffset: 0 # time offset. (min)
deadTime: [] # deadtime (ns). If it's empty, deadtime correction is disabled.
bgBins: [1150, 1250] # [start index, stop index] for background correction
nPretrigger: 0 # if nPretrigger < 0, move signal up
bgCorFile: '' # data file of dark measurement results
overlapFile: ''
AW:
lidarNo: 7 # lidar number. (see ./docs/lidarList.md)
chTag: ['532p', '532s']
fullOverlapHeight: 400 # minimum height with complete overlap. (m)
hOffset: 0 # height offset. (m)
tOffset: 0 # time offset. (min)
deadTime: [] # deadtime (ns). If it's empty, deadtime correction is disabled.
bgBins: [1600, 1700] # [start index, stop index] for background correction
nPretrigger: 0 # if nPretrigger < 0, move signal up
bgCorFile: '' # data file of dark measurement results
overlapFile: ''
ZK:

```

```
lidarNo: 2 # lidar number. (see ./docs/lidarList.md)
chTag: ['532p', '532s']
fullOverlapHeight: 400 # minimum height with complete overlap. (m)
hOffset: 0 # height offset. (m)
tOffset: 0 # time offset. (min)
deadTime: [] # deadtime (ns). If it's empty, deadtime correction is disabled.
bgBins: [1800, 1900] # [start index, stop index] for background correction
nPretrigger: 0 # if nPretrigger < 0, move signal up
bgCorFile: '' # data file of dark measurement results
overlapFile: ''
```

GB:

```
lidarNo: 8 # lidar number. (see ./docs/lidarList.md)
chTag: ['532p', '532s']
fullOverlapHeight: 400 # minimum height with complete overlap. (m)
hOffset: 0 # height offset. (m)
tOffset: 0 # time offset. (min)
deadTime: [] # deadtime (ns). If it's empty, deadtime correction is disabled.
bgBins: [1200, 1300] # [start index, stop index] for background correction
nPretrigger: 0 # if nPretrigger < 0, move signal up
bgCorFile: '' # data file of dark measurement results
overlapFile: ''
```

SY:

```
lidarNo: 14 # lidar number. (see ./docs/lidarList.md)
chTag: ['532p', '532s', '355e', '607', '1064e']
fullOverlapHeight: 400 # minimum height with complete overlap. (m)
hOffset: 0 # height offset. (m)
tOffset: 0 # time offset. (min)
deadTime: [] # deadtime (ns). If it's empty, deadtime correction is disabled.
bgBins: [1400, 1500] # [start index, stop index] for background correction
nPretrigger: 0 # if nPretrigger < 0, move signal up
bgCorFile: '' # data file of dark measurement results
overlapFile: ''
```

SS:

```
lidarNo: 7 # lidar number. (see ./docs/lidarList.md)
chTag: ['532p', '532s']
fullOverlapHeight: 400 # minimum height with complete overlap. (m)
hOffset: 0 # height offset. (m)
tOffset: 0 # time offset. (min)
deadTime: [] # deadtime (ns). If it's empty, deadtime correction is disabled.
bgBins: [900, 950] # [start index, stop index] for background correction
nPretrigger: 0 # if nPretrigger < 0, move signal up
bgCorFile: '' # data file of dark measurement results
overlapFile: ''
```

L7:

```
lidarNo: 9 # lidar number. (see ./docs/lidarList.md)
chTag: ['355p', '355s', '532p', '532s', '387', '407', '607', '1064e']
fullOverlapHeight: 400 # minimum height with complete overlap. (m)
hOffset: 0 # height offset. (m)
```

```

tOffset: 0 # time offset. (min)
deadTime: [] # deadtime (ns). If it's empty, deadtime correction is disabled.
bgBins: [1500, 1600] # [start index, stop index] for background correction
nPretrigger: 0 # if nPretrigger < 0, move signal up
bgCorFile: '' # data file of dark measurement results
overlapFile: ''

```

每个配置参数的说明可以参考[配置文件说明.pdf](#)。

接下来可以配置互对比中的距离修正信号对比的相关配置，这些配置主要包括使用哪些雷达，如何从不同通道数据中组合弹性信号，并且控制输出图形的横纵坐标范围

```

figVisible: 'on' # whether display figures
flagRangeCmp: false
flagRCSCmp: true
flagVDRCmp: false
flagFernaldCmp: false
flagRamanCmp: false
RCSCmpCfg: # 532 p
  LidarList: ['REAL', 'SY', 'L7', 'SS', 'ZK', 'GB', 'AW']
  sigCompose:
    - [0, 1, 0, 0, 0, 0] # first lidar
    - [1, 0, 0, 0, 0] # second lidar
    - [0, 0, 1, 0, 0, 0, 0, 0] # second lidar
    - [0, 1] # second lidar
    - [1, 0] # second lidar
    - [1, 0] # second lidar
    - [1, 0] # second lidar
  tRange: '2021-09-27 19:00:00 ~ 2021-09-27 19:30:00'
  normRange: [3500, 4500]
  hChkRange:
    - [500, 2000]
    - [2000, 5000]
  maxDev:
    - 10
    - 20
  smoothwindow:
    - [0, 1000, 100] # Piecewise smoothing: [bottom height, top height, window length]
    - [1000, 10000, 300]
    - [10000, 20000, 600]
  hRange: [0, 14000]
  sigRange: [1e10, 1e15]

```

其中 flagRangeCmp 代表距离测量精度对比，flagRCSCmp 代表距离修正信号对比，flagVDRCmp 代表体退偏结果对比，flagFernaldCmp 代表 Fernald 反演结果对比，flagRamanCmp 代表 Raman 反演结果对比。距离修正信号对比的配置参数在 RCSCmpCfg 对应的

列表下，其中 LidarList 代表需要进行互对比的雷达列表。sigCompose 表示每台雷达如何组合成对应的距离修正信号，这里每条对应通道信号组合时的系数，比如[0, 1]表示第一个通道信号的系数为 0，第二个通道信号的系数为 1。tRange 表示选取的信号时间段。normRange 表示信号归一化的高度范围，单位为米。hChkRange 表示信号偏移评估的高度范围，以上示例中的高度范围为激光雷达标定标准建议的范围。maxDev 表示每个评估高度范围内允许的最大偏差。Smoothwindow 为分段平滑窗，每条信息为一个滑动平滑窗，第一个数值为滑动平滑窗的起始高度。第二个为滑动平滑窗的终止高度，最后一个为滑动平滑窗的窗长。hRange 为显示结果的高度范围。sigRange 为信号的显示范围。